Mustafa Furkan Ceylan

Researcher in AI Security, Edge AI & IoT Cybersecurity

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PROFILE

I am a researcher working at the intersection of artificial intelligence, cybersecurity and edge computing. My doctoral research focuses on AI-based intrusion detection for resource-constrained IoT environments, with particular interests in lightweight models, explainability and human-supervised security systems.

I am a Research Assistant at Balıkesir University and a PhD researcher in Computer Engineering at Dokuz Eylül University. Before academia, I worked for nearly three years in software test automation at NETAŞ. I welcome research collaborations, selected AI projects, technical training and interdisciplinary work.

RESEARCH INTERESTS

Artificial Intelligence for Cybersecurity · Intrusion Detection Systems · Internet of Things · Edge AI and Lightweight Models · Explainable Artificial Intelligence · Network Security · Secure Cyber-Physical Systems · AI for Medicine

EXPERIENCE & EDUCATION

PhD in Computer Engineering
2025-Present · Dokuz Eylül University
English-taught doctoral programme

Research Assistant
Sep 2023-Present · Balıkesir University
Department of Computer Engineering

Visiting Researcher
Jul-Sep 2025 · Flanders Research Institute for Agriculture, Fisheries and Food (ILVO), Belgium
Animal Sciences - Artificial Intelligence

MSc in Computer and Informatics Engineering
2022-Nov 2024 · Balıkesir University
Thesis: Detection of ARP Spoofing Using Machine Learning

Erasmus+ MSc Study Mobility
Feb-Jul 2023 · Częstochowa University of Technology, Poland
International study mobility

Senior Software Test Automation Engineer
Dec 2020-Sep 2023 · NETAŞ Telecommunications
Web and mobile test automation, performance testing and mentoring

BSc in Electrical and Electronics Engineering
2015-Sep 2020 · Uludağ University
Faculty of Engineering

FUNDED & SELECTED PROJECTS

AI-Based Multi-Agent Intrusion Detection System
TÜBİTAK 1002 · Scholarship Researcher · Sep 2026-Aug 2027
A funded research project investigating explainable edge intrusion detection and multi-agent support for actionable security analysis in IoT environments.

Journey of Beliefs: Interactive Mapping of the History of Religions
TÜBİTAK 3005 · Researcher · Dec 2025-2027
An interdisciplinary project developing thematic, interactive and open-access maps for the visualization of the history of world religions.

AISC - AI and Security Research Group

Balıkesir University · Founding Researcher · 2026-Present

A growing student research community focused on artificial intelligence, cybersecurity and applied research through weekly training, mentoring and collaborative projects.

Detection of ARP Spoofing Using Machine Learning

MSc Thesis · MSc Researcher · Completed Nov 2024

A network-security study using machine-learning and deep-learning models to detect ARP spoofing from packet-derived features in IoT traffic.

SELECTED PUBLICATIONS

1. AI Agent-Assisted Lightweight Intrusion Detection for IoT Devices

Doğancan Karakoç, Gökhan Dalkılıç, Mustafa Furkan Ceylan

2026 8th International Congress on Human-Computer Interaction, Optimization and Robotic Applications (ICHORA) (2026) · DOI: 10.1109/ICHORA69329.2026.11537202

2. Stage Classification of Alzheimer's Disease MRI Data via Deep Neural Network Architectures

Aslıhan Güven, Abdurrahim Hüseyin Ezirmik, Mustafa Furkan Ceylan, Fatih Aydın

Balıkesir Journal of Engineering Sciences and Research, 8(1), 93-103 (2026) · DOI: 10.46387/bjesr.1869200

3. Comparative Evaluation of Cloud-Edge Security Architectures for DDoS Detection

Mustafa Furkan Ceylan, Gökhan Dalkılıç, Mehmet Hilal Özcanhan

2025 18th International Conference on Information Security and Cryptology (ISCTürkiye), 122-127 (2025) · DOI: 10.1109/ISCTrkiye68593.2025.11224857

4. DDoS Detection in Network Traffic Using LightGBM: A Study on the CICIDS2018 Dataset

Mustafa Furkan Ceylan, Selçuk Kavut, Faruk Karadeniz

9th International Symposium on Innovative Approaches in Smart Technologies (ISAS) (2025) · DOI: 10.1109/ISAS66241.2025.11101816

5. Penetration Testing: A Review of Standards, Processes and Deception Techniques

Mustafa Furkan Ceylan, Selçuk Kavut

Journal of Computer Science and Engineering, 18(1), 64-85 (2025) · DOI: 10.54525/bbmd.1652127

TECHNICAL SKILLS

Python · Java · Machine Learning · Deep Learning · Scikit-learn · TensorFlow / Keras · Data Analysis · Git / GitHub · Docker · Linux · Raspberry Pi · TensorFlow Lite · TShark / Wireshark · Selenium · Appium · JMeter · TestNG

TEACHING & MENTORING

Electronic Circuits Laboratory - Undergraduate laboratory teaching
Digital Design Laboratory - Undergraduate laboratory teaching

AISC - AI and Security Research Group · Founding Researcher

Weekly research and training meetings · AI and cybersecurity education · Literature review and research methodology · Student-led research projects · Dataset analysis and AI model development

TALKS & ACADEMIC ACTIVITIES

Detection of ARP Spoofing Using Machine Learning

Research Presentation · Mälardalen University, Sweden · Dec 2024

PROFILES

Google Scholar: https://scholar.google.com/citations?user=JmfYJIIAAAAJ&hl=en · ORCID: https://orcid.org/0009-0005-5609-395X · GitHub: https://github.com/MFCEYLAN · LinkedIn: https://www.linkedin.com/in/mustafafurkanceylan/ · YouTube: https://www.youtube.com/@mustafafurkanceylan/featured